UPAIR: Universal Prompted Axial In-context Estimator for 5G New Radio Uplink

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***Abstract*—This paper proposes UPAIR (Universal Prompted Axial In-context estimatoR), a prompted channel-estimation front-end for 5G New Radio (NR). The estimator plugs into the standard NR uplink receiver chain, leaving the rest of the chain unchanged. Starting from a least-squares (LS) estimate on the demodulation reference signals (DMRS), UPAIR forms a slot-level prompt that conditions feature-wise linear modulation (FiLM) axial refinement blocks, and outputs a channel estimate together with an uncertainty map consumed by the downstream linear minimum mean-square error (LMMSE) detector. We evaluate UPAIR against classical channel estimators in three scenarios: a clean reference case, a mild phase-distortion case, and a DMRS-rich mild phase-distortion case. Across these scenarios at BLER of 10^{-2}, UPAIR achieves gains of 0.80 dB and 1.18 dB compared to the strongest classical baseline in the clean and mild phase-distortion settings, respectively. In the latter, UPAIR remains only about 0.55 dB from the perfect-CSI benchmark. In the DMRS-rich mild phase-distortion setting, UPAIR is essentially tied with the strongest classical baseline. Over the full evaluated SNR grid, UPAIR also yields the lowest practical channel-estimation error in all three scenarios.**

I. Introduction

Reliable channel estimation remains a central requirement in 5G New Radio (NR) because detection and decoding depends directly on the channel state information (CSI) extracted from the demodulation reference signals (DMRS) defined by the standard [1], [2]. Classical OFDM studies established techniques that still underlie practical receivers, including single-value decomposition-based linear minimum mean squared error (LMMSE) estimation [3], robust MMSE estimation for dispersive channels [4], and pilot-aided maximum likelihood and Bayesian MMSE [5].

A related challenge is oscillator impairment: OFDM sensitivity to carrier-frequency offset and Wiener phase noise [6], and the resulting common phase error [7], remain relevant whenever the density of pilot symbols is limited and the receiver must interpolate while also tracking phase variations.

Deep learning (DL) has provided an alternative to purely model-based estimation [8]–[10], with OFDM-specific studies demonstrating gains in channel estimation and signal detection under distortion [11]. In this direction, axial self-attention factorizes attention along the time and frequency axes, matching the separable structure of OFDM and reducing complexity [12], while more recent efforts target standard-compliant deployments, including a neural receiver for 5G NR multi-user MIMO [13] and real-time operation under latency constraints [14].

A recent trend is prompt-based or in-context wireless inference. In [15], transformers were shown to asymptotically achieve optimal in-context estimation for a class of wireless estimation problems. Further work then extended this idea to decision-feedback symbol detection [16] and to a multi-task OFDM transformer for channel estimation and symbol detection without task-specific fine-tuning [17]. Together, these results show that adaptation can come from the current input and context, rather than from retraining during deployment.

In parallel, large-model ideas also began to influence wireless channel processing. Recent examples include LLM-based multi-task wireless processing [18], generative pre-training and large-model channel modeling [19], [20], direct use of pre-trained vision models for CSI-related tasks [21], and context-conditioned estimators based on adaptive transformers and feature-wise linear modulation (FiLM)-guided diffusion [22], [23]. Overall, these works show that prompting, context conditioning, and large models are relevant not only to broader wireless learning tasks but also to channel estimation itself.

From a practical standpoint, an important unsolved design issue remains attractive: the standard-compliant 5G NR uplink chain remains unchanged except for the channel-estimation front-end, preserving compatibility with the conventional downstream detection and decoding chain and making the perfect-CSI curve a meaningful benchmark. The present paper follows this direction and proposes Universal Prompted Axial In-context estimatoR (UPAIR), a slot-conditioned channel-estimation front-end for 5G NR. Starting from an LS reference, it maps current-slot observations to a feature tensor, forms a slot prompt by emodulation reference signals (DMRS)-masked averaging of the projected features followed by a small multilayer perceptron (MLP), and refines the estimate through FiLM-conditioned axial processing [24], [25].

The rest of the paper presents the system model, baseline receivers, and numerical results showing the proposed front-end outperforms practical baselines and approaches the perfect-CSI benchmark within a standard-compliant framework.

II. System Model and Receiver Objective

We consider a slot comprising T OFDM symbols and F active subcarriers. For one transmit layer and N_r receive antennas, the received frequency-domain sample on resource element (t, f) and antenna r follows the standard OFDM model with residual common phase error [6], [7],

$$y_{t,f,r} = e^{j\phi_t} h_{t,f,r} x_{t,f} + n_{t,f,r}, \tag{1}$$

where $x_{t,f}$ is the transmitted physical uplink shared channel (PUSCH) symbol, $h_{t,f,r}$ is the propagation channel, ϕ_t is a residual common phase error term shared by all subcarriers of OFDM symbol t, and $n_{t,f,r}$ is additive white Gaussian noise. Collecting $y_{t,f,r}$ over all (t, f, r) yields the received resource grid $\mathbf{Y} \in \mathbb{C}^{T\times F\times N_r}$. The set of DMRS positions is denoted by $\mathcal{P}$ and the set of data resource elements by $\mathcal{D}$. The binary pilot mask $\mathbf{M}_{\mathcal{P}} \in \{0, 1\}^{T\times F}$ satisfies $[\mathbf{M}_{\mathcal{P}}]_{t,f} = 1$ for $(t, f) \in \mathcal{P}$ and 0 otherwise. We use the Wiener phase error model, i.e., the received grid is rotated by a symbol-wise phase-error model that combines a common phase offset, a linear drift across OFDM symbols, and a random-walk term, namely,

$$\phi_t = \phi_0 + \alpha\,\bar{t} + \sum_{\tau=0}^{t} w_\tau, \tag{2}$$

where $\bar{t} = (t-(T-1)/2)/(T-1)$ is the normalized centered symbol index, ϕ_0 is the random initial phase, α is the slot-level slope coefficient, and $w_\tau \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_w^2)$ are independent Gaussian increments.

The standard LS estimator uses the DMRS observations and interpolates them over the full resource grid. Denoting the interpolated LS estimate as $\widehat{\mathbf{H}}^{\text{LS}}$ and its element-wise error variance as $\mathbf{V}^{\text{LS}}$, the estimator problem can be stated as follows: from the current slot observations $\mathbf{Y}$, the noise level N_0, the pilot mask $\mathbf{M}_{\mathcal{P}}$, and the LS estimate, construct a refined channel estimate $\widehat{\mathbf{H}}$ and a refined uncertainty map $\widehat{\mathbf{V}}$ that can be consumed directly by the downstream detector; in this paper the downstream detector is the standard LMMSE detector, but the interface is not specific to that detector.

A key design goal is slotwise adaptation without gradient updates at inference. During evaluation, all stages remain feed-forward, and UPAIR can exploit only information from the current DMRS pattern and the current received grid.

III. UPAIR Channel-Estimation front-end

Figure 1 summarizes the UPAIR pipeline. The estimator takes the current slot observations $(\mathbf{Y}, N_0, \mathbf{M}_{\mathcal{P}})$ and the LS reference $(\widehat{\mathbf{H}}^{\text{LS}}, \mathbf{V}^{\text{LS}})$ as inputs and produces a refined channel estimate $\widehat{\mathbf{H}}$ and an uncertainty map $\widehat{\mathbf{V}}$. It proceeds in four steps: (i) feature construction, (ii) slot prompt extraction, (iii) FiLM-conditioned axial refinement, and (iv) output head projections. Each step is detailed below.

A. Feature construction

The estimator first converts the available slot information into a single real-valued feature tensor. For $\mathbf{A} \in \mathbb{C}^{T\times F\times N_r}$, define the real–imaginary stacking operator $\mathcal{R}(\mathbf{A}) \triangleq \text{concat}_{\text{feat}}(\Re\{\mathbf{A}\}, \Im\{\mathbf{A}\}) \in \mathbb{R}^{T\times F\times 2N_r}$, which stacks the real and imaginary parts of the N_r receive-antenna slices along the feature axis. Then,

$$\begin{aligned}\mathbf{F} &= \text{concat}_{\text{feat}}\Big(\mathcal{R}(\widehat{\mathbf{H}}^{\text{LS}}), \mathcal{R}(\mathbf{Y}), \overline{\mathbf{V}}^{\text{LS}}, \mathbf{N}_0, \mathbf{M}_{\mathcal{P}}^{\text{map}}\Big) \\ &\in \mathbb{R}^{T\times F\times(4N_r+3)}.\end{aligned} \tag{3}$$

Here, $\text{concat}_{\text{feat}}(\cdot)$ denotes concatenation along the third, feature dimension, $\mathbf{N}_0 = N_0\mathbf{1}_{T\times F\times 1}$, and $\mathbf{M}_{\mathcal{P}}^{\text{map}}$ is the $T\times F$ pilot mask viewed as a $T \times F \times 1$ map. Thus the feature dimension is $4N_r + 3$, e.g., 11 for $N_r = 2$.

A 1×1 convolutional projection (i.e., a learned linear map applied independently at every (t, f)) then projects the per-(t, f) feature vector to dimension d. That is,

$$\big[\mathbf{Z}^{(0)}\big]_{t,f,:} = \mathbf{W}_{\text{con}}\big[\mathbf{F}\big]_{t,f,:} + \mathbf{b}_{\text{con}}, \qquad \mathbf{Z}^{(0)} \in \mathbb{R}^{T\times F\times d}, \tag{4}$$

where $\mathbf{W}_{\text{con}} \in \mathbb{R}^{d\times(4N_r+3)}$ and $\mathbf{b}_{\text{con}} \in \mathbb{R}^d$. The feature tensor $\mathbf{Z}^{(0)}$ is the shared starting point for both the slot-prompt extraction and the subsequent axial refinement.

B. Slot-prompt extraction

The slot prompt $\mathbf{p} \in \mathbb{R}^d$ is a compact vector that summarizes the channel conditions of the current slot using only the reliable DMRS positions. It is obtained by averaging the projected features over all pilot locations $\mathcal{P}$ and passing the result through a two-layer MLP g_ψ. Particularly,

$$\mathbf{p} = g_\psi\left(\frac{1}{|\mathcal{P}|}\sum_{(t,f)\in\mathcal{P}} \mathbf{Z}^{(0)}_{t,f,:}\right). \tag{5}$$

The DMRS-masked average uses information only from positions where the channel is directly observed, and avoids noisy data resource elements. The prompt is computed once per slot and held fixed through the refinement phase.

C. FiLM-conditioned axial refinement

1) FiLM conditioning: At the beginning of each sub-stage inside every axial block, the current block state $\mathbf{X}_i \in \mathbb{R}^{T\times F\times d}$ is normalized and modulated by the slot prompt. First, the prompt is projected to a gain vector $\boldsymbol{\gamma}(\mathbf{p}) \in \mathbb{R}^d$ and a bias vector as $\boldsymbol{\beta}(\mathbf{p}) \in \mathbb{R}^d$

$$\big[\boldsymbol{\gamma}(\mathbf{p}),\ \boldsymbol{\beta}(\mathbf{p})\big] = \text{split}(\mathbf{W}_p\mathbf{p} + \mathbf{b}_p)\,, \tag{6}$$

where $\mathbf{W}_p \in \mathbb{R}^{2d\times d}$ and $\mathbf{b}_p \in \mathbb{R}^{2d}$ are learned parameters, and $\text{split}(\cdot)$ partitions the $2d$-dimensional output into two d-dimensional halves. The FiLM-conditioned tensor is then formed by applying these vectors uniformly to every resource element (t, f) according to

$$\widetilde{\mathbf{X}}_{i,t,f,:} = \text{LN}\big(\mathbf{X}_{i,t,f,:}\big) \odot \big(\mathbf{1} + \boldsymbol{\gamma}(\mathbf{p})\big) + \boldsymbol{\beta}(\mathbf{p}), \tag{7}$$

where $\text{LN}(\cdot)$ denotes layer normalization applied independently at each (t, f) position, and $\odot$ is element-wise multiplication. Because $\boldsymbol{\gamma}$ and $\boldsymbol{\beta}$ are computed from $\mathbf{p}$ and broadcast over all (t, f), the same slot-level modulation is applied uniformly across the entire resource grid. The resulting tensor $\widetilde{\mathbf{X}}_i$ is the input to the ith sub-stage of the block. Figure 2 illustrates this operation.

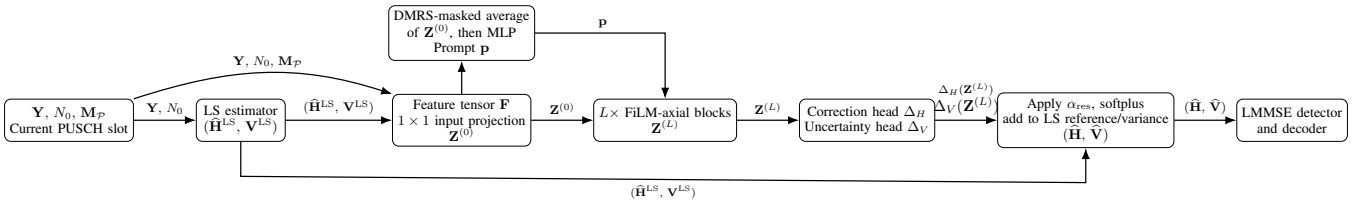

Fig. 1. UPAIR receiver flow. The LS output and current slot observations are mapped to $\mathbf{F}$, projected to $\mathbf{Z}^{(0)}$, summarized over DMRS positions to form the prompt $\mathbf{p}$, refined by L FiLM-axial blocks, and converted to $(\widehat{\mathbf{H}}, \widehat{\mathbf{V}})$ for the standard LMMSE detector. Fig. 3 details one axial block, and Fig. 2 isolates the FiLM step.

2) Axial block structure: Each FiLM-conditioned axial block transforms the input tensor $\mathbf{X}_0 = \mathbf{Z}^{(\ell)} \in \mathbb{R}^{T\times F\times d}$ through four sequential sub-stages, each with a residual skip connection. The term *axial* refers to the fact that self-attention is applied one-dimensionally along a single axis of the resource grid at a time, either across the F subcarriers within a fixed OFDM symbol, or across the T OFDM symbols at a fixed subcarrier (rather than jointly over the full two-dimensional grid). This factorization not only matches the separable time-frequency structure of OFDM, but also reduces the computational complexity from $O((TF)^2)$ to $O(TF^2+FT^2)$. The four sub-stages are as follows (see also Fig. 3).

Sub-stage 1 — Frequency self-attention. The FiLM-conditioned slice $\widetilde{\mathbf{X}}_{0,t,:,:} \in \mathbb{R}^{F\times d}$ at each OFDM symbol t is processed by a multi-head self-attention (MHA) module applied independently across the F subcarriers. Let $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{F\times d}$ denote the query, key, and value projections of this slice. The scaled dot-product attention output for symbol t is

$$\mathcal{A}_f^t = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d/H}}\right)\mathbf{V} \ \in \mathbb{R}^{F\times d}, \tag{8}$$

where H is the number of attention heads and d/H is the per-head dimension. After processing all T symbols independently, the outputs are stacked back to the full grid and added to $\mathbf{X}_0$ through a residual connection $\mathbf{X}_1 = \mathbf{X}_0 + \mathrm{Drop}_\rho(\mathcal{A}_f(\widetilde{\mathbf{X}}_0))$, where $\mathrm{Drop}_\rho(\cdot)$ denotes dropout with rate ρ (active only during training, identity at inference), and $\mathcal{A}_f(\widetilde{\mathbf{X}}_0)$ stacks the per-symbol outputs $\mathcal{A}_f^t$ over all t. This sub-stage allows every subcarrier to attend to all other subcarriers within the same symbol, capturing frequency-domain channel correlations.

Sub-stage 2 — Time self-attention. The same MHA operation is now applied along the time axis. The FiLM-conditioned slice $\widetilde{\mathbf{X}}_{1,:,f,:} \in \mathbb{R}^{T\times d}$ at each subcarrier f is processed independently, yielding the per-subcarrier time-attention output $\mathcal{A}_t^f \in \mathbb{R}^{T\times d}$. After stacking over all f, we have $\mathbf{X}_2 = \mathbf{X}_1 + \mathrm{Drop}_\rho(\mathcal{A}_t(\widetilde{\mathbf{X}}_1))$. This sub-stage lets every OFDM symbol attend to all other symbols at the same subcarrier, capturing temporal channel variations.

Sub-stage 3 — Depthwise-pointwise convolution. To capture short-range local structure in the time-frequency grid, this sub-stage applies a depth-separable convolution to $\widetilde{\mathbf{X}}_2$. A depthwise 3×3 convolution (DWConv) processes each of the d feature maps independently, applying one distinct 3×3 spatial kernel k_c to each feature map.

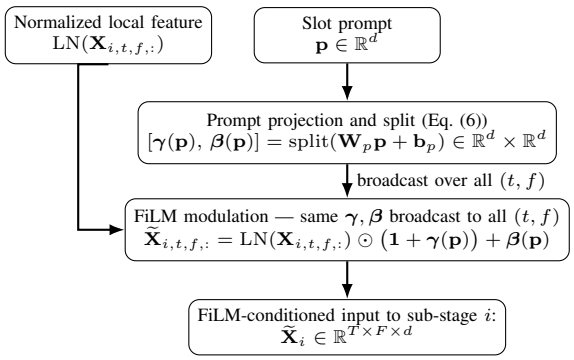

Fig. 2. FiLM conditioning applied at the ith sub-stage of each axial block.

A pointwise 1×1 convolution (PWConv) then mixes the d features independently at each resource element. Let $\mathbf{U}_{t,f,:} \in \mathbb{R}^d$ denote the local feature vector at (t, f). The operation is

$$[\mathrm{PWConv}(\mathbf{U})]_{t,f,:} = \mathbf{W}\mathbf{U}_{t,f,:} + \mathbf{b}, \tag{9}$$

where $\mathbf{W} \in \mathbb{R}^{d\times d}$ and $\mathbf{b} \in \mathbb{R}^d$ are shared across all (t, f). The branch output is $\mathcal{C}(\widetilde{\mathbf{X}}_2) = \mathrm{Drop}_\rho(\mathrm{GELU}(\mathrm{PWConv}(\mathrm{DWConv}(\widetilde{\mathbf{X}}_2))))$, and the residual update is $\mathbf{X}_3 = \mathbf{X}_2 + \mathcal{C}(\widetilde{\mathbf{X}}_2)$. Together, DWConv and PWConv form a depthwise-separable convolution: DWConv captures local structure, and PWConv mixes features per position.

Sub-stage 4 — Pointwise MLP. A small MLP is applied independently at every (t, f) position to mix features. It expands the d-dimensional feature vector to dimension rd via a fully connected layer FC_1 (expansion ratio r), applies GELU and dropout, then projects back to d via FC_2 followed by another dropout. This can be written compactly as

$$\begin{aligned}&\left[\mathcal{M}(\widetilde{\mathbf{X}}_3)\right]_{t,f,:} = \\ &\mathrm{Drop}_\rho\Big(\mathbf{W}_2\,\mathrm{Drop}_\rho\Big(\mathrm{GELU}\Big(\mathbf{W}_1\widetilde{\mathbf{X}}_{3,t,f,:} + \mathbf{b}_1\Big)\Big) + \mathbf{b}_2\Big),\end{aligned} \tag{10}$$

where FC_1 and FC_2 use weight matrices $\mathbf{W}_1 \in \mathbb{R}^{rd\times d}$ and $\mathbf{W}_2 \in \mathbb{R}^{d\times rd}$, and bias vectors $\mathbf{b}_1 \in \mathbb{R}^{rd}$ and $\mathbf{b}_2 \in \mathbb{R}^d$ respectively, shared across all (t, f). The residual update gives the block output $\mathbf{X}_4 = \mathbf{Z}^{(\ell+1)} = \mathbf{X}_3 + \mathcal{M}(\widetilde{\mathbf{X}}_3)$. Stacking L such blocks yields the final representation $\mathbf{Z}^{(L)} = \mathcal{B}_{L-1}\big(\cdots\mathcal{B}_0\big(\mathbf{Z}^{(0)};\mathbf{p}\big)\cdots;\mathbf{p}\big)$, where $\mathcal{B}_\ell(\cdot\,;\mathbf{p})$ denotes the ℓth block conditioned on the slot prompt $\mathbf{p}$.

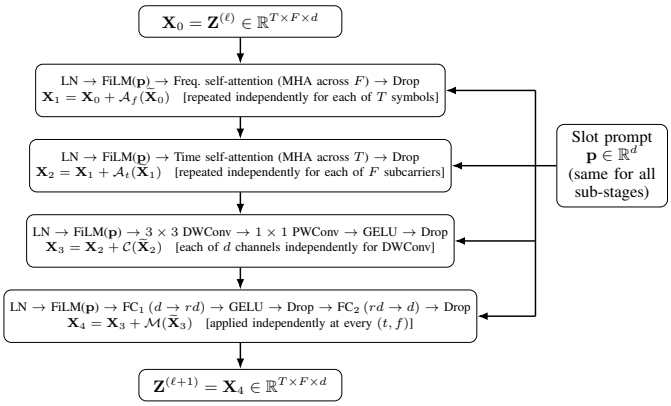

Fig. 3. One FiLM-conditioned axial block $\mathcal{B}_\ell$.

D. Correction and uncertainty output heads

After the last axial block, the final tensor $\mathbf{Z}^{(L)} \in \mathbb{R}^{T\times F\times d}$ is projected by two parallel 1×1 convolutional heads. Both heads are pointwise maps (i.e., they operate independently at every (t, f), sharing the same weights across the grid).

Correction head Δ_H: A 1×1 convolution projects $\mathbf{Z}^{(L)}$ from feature dimension d to $2N_r$ real-valued outputs, grouped into real and imaginary parts to form a complex correction $\Delta_H(\mathbf{Z}^{(L)}) \in \mathbb{C}^{T\times F\times N_r}$. The refined channel estimate is

$$\widehat{\mathbf{H}} = \widehat{\mathbf{H}}^{\mathrm{LS}} + \alpha_{\mathrm{res}}\, \Delta_H\Big(\mathbf{Z}^{(L)}\Big), \tag{11}$$

where α_{res} is a fixed chosen correction scale factor.

Uncertainty head Δ_V: A separate 1×1 convolution maps d to N_r real values. The refined uncertainty map is

$$\widehat{\mathbf{V}} = \mathbf{V}^{\mathrm{LS}} + \mathrm{softplus}\Big(\Delta_V\Big(\mathbf{Z}^{(L)}\Big)\Big), \tag{12}$$

where $\mathrm{softplus}(x) = \log(1 + e^x) \geq 0$. The uncertainty head selectively increases the estimated variance in unreliable regions, which directly affects the downstream LMMSE detector. The pair $(\widehat{\mathbf{H}}, \widehat{\mathbf{V}})$ is passed to the standard downstream LMMSE detector without any further modification.

E. Training objective

During training, the estimator is trained end-to-end on freshly generated PUSCH slots. For each slot, the ground-truth channel tensor $\mathbf{H}$ is available from the channel simulator. The loss combines a heteroscedastic Gaussian negative log-likelihood (NLL) term with an normalized MSE (NMSE) regularizer as

$$\mathcal{L} = \underbrace{\mathrm{mean}\left(\frac{\left|\mathbf{H} - \widehat{\mathbf{H}}\right|^2}{\widehat{\mathbf{V}} + \varepsilon} + \log\Big(\widehat{\mathbf{V}} + \varepsilon\Big)\right)}_{\text{heteroscedastic NLL}} + \lambda\, \underbrace{\frac{\mathrm{mean}\left(\left|\mathbf{H} - \widehat{\mathbf{H}}\right|^2\right)}{\mathrm{mean}\big(|\mathbf{H}|^2\big) + \varepsilon}}_{\text{NMSE regularizer}}, \tag{13}$$

where $\varepsilon > 0$ is a numerical-stability constant and λ weights the NMSE term. The NLL term jointly trains the correction head and the uncertainty head: it encourages $\widehat{\mathbf{H}}$ to approach $\mathbf{H}$ while $\widehat{\mathbf{V}}$ reflects the actual error magnitude. Training uses an adaptive optimizer with weight decay, gradient clipping, and checkpointing of the best validation model.

Remark.: The refined pair $(\widehat{\mathbf{H}}, \widehat{\mathbf{V}})$ is consumed directly by the unchanged downstream LMMSE detector. In the single-layer setting considered in this paper, for each resource element (t, f), the detector uses $\widehat{H}_{t,f,r}$ as the channel estimate on receive antenna r and uses $\widehat{V}_{t,f,r}$ as the corresponding CSI-uncertainty variance. Hence, $\widehat{\mathbf{V}}$ enters the detector as an additional diagonal term to the noise variance N_0, so that antennas or resource elements with larger estimated uncertainty are trusted less. The resulting post-equalized symbol can be written as

$$\widehat{x}_{t,f} = \frac{\displaystyle\sum_{r=1}^{N_r} \frac{\widehat{H}^*_{t,f,r}}{N_0 + \widehat{V}_{t,f,r}}\, y_{t,f,r}}{\displaystyle\sum_{r=1}^{N_r} \frac{\left|\widehat{H}_{t,f,r}\right|^2}{N_0 + \widehat{V}_{t,f,r}}}, \tag{14}$$

with corresponding effective post-equalization noise variance

$$\sigma^2_{t,f} = \left(\sum_{r=1}^{N_r} \frac{\left|\widehat{H}_{t,f,r}\right|^2}{N_0 + \widehat{V}_{t,f,r}}\right)^{-1}. \tag{15}$$

The pair $(\widehat{x}_{t,f}, \sigma^2_{t,f})$ is then passed to the standard soft demapper to produce the bit LLRs.

IV. Baselines

The practical baselines we used are the standard LS, LS + 2D LMMSE, and an LS baseline augmented with decision-directed common-phase-error correction [26], [27]. The LS + 2D LMMSE receiver replaces the default LS interpolation by a separable LMMSE interpolator built from empirical covariance matrices $\mathbf{R}_t$, $\mathbf{R}_f$, and $\mathbf{R}_s$, estimated offline from generated channel realizations, with interpolation order frequency–time–space.

For the DD-CPE baseline, the iteration is initialized by $\widehat{\mathbf{H}}^{(0)} = \widehat{\mathbf{H}}^{\mathrm{LS}}$. At iteration i, the equalized symbol is

$$z^{(i)}_{t,f} = \frac{\sum_{r=1}^{N_r} \big(\widehat{h}^{(i)}_{t,f,r}\big)^* y_{t,f,r}}{\sum_{r=1}^{N_r} \left|\widehat{h}^{(i)}_{t,f,r}\right|^2 + N_0}. \tag{16}$$

Using the hard decision $\widehat{s}^{(i)}_{t,f}$, one residual common phase per OFDM symbol is estimated as

$$\widehat{\theta}^{(i)}_t = \angle\left(\sum_{f\in\mathcal{D}_t} \left(\sum_{r=1}^{N_r} \left|\widehat{h}^{(i)}_{t,f,r}\right|^2\right)^{\eta} z^{(i)}_{t,f} \left(\widehat{s}^{(i)}_{t,f}\right)^*\right), \tag{17}$$

where $\mathcal{D}_t = \{f : (t, f) \in \mathcal{D}\}$ and η is a weighting exponent. The estimated phase sequence is then shifted to be zero at the DMRS symbol, smoothed, clipped, and used to update $\widehat{h}^{(i+1)}_{t,f,r} = e^{j\widehat{\theta}^{(i)}_t}\, \widehat{h}^{(i)}_{t,f,r}$. In this paper, two DD-CPE iterations are used.

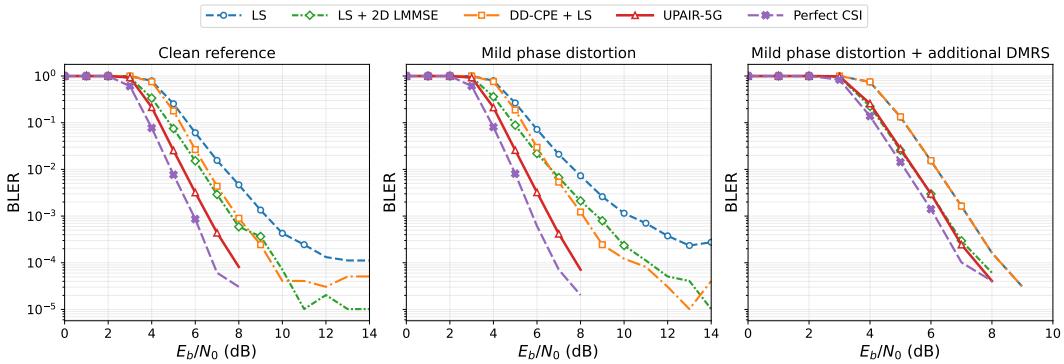

Fig. 4. Cross-scenario BLER comparison for the clean reference, mild phase-distortion, and DMRS-rich mild phase settings.

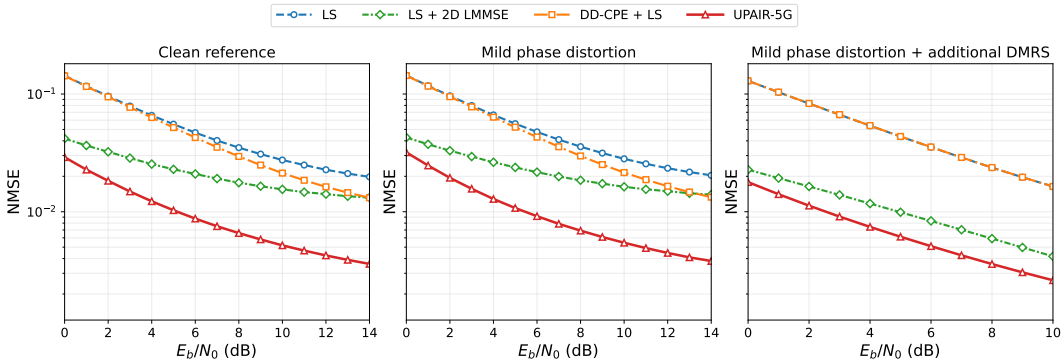

Fig. 5. Cross-scenario NMSE comparison for the clean reference, mild phase-distortion, and DMRS-rich mild phase settings.

V. Simulation Results

Our simulations are based on the open-source package, Sionna [28]. All three evaluated scenarios share the same NR numerology and receiver chain instantiated in Sionna: carrier frequency 3.5 GHz, subcarrier spacing 30 kHz, PUSCH mapping type A, $T = 14$ OFDM symbols, normal cyclic prefix, 16 resource blocks, one layer, one antenna port, and 16-QAM with MCS index 15. The propagation channel is uplink CDL-C [29] with 300 ns delay spread, one transmit antenna, two receive antennas, and user speed sampled uniformly from 30 to 60 km/h. The DMRS uses configuration type 1, length 1, and type-A position 2.

Table I summarizes the three scenarios. In the two impaired scenarios, the phase process in (2) is enabled. During training, ϕ_0 has standard deviation $\sigma_\phi = 0.012$ rad, α is sampled uniformly from $[-0.08, 0.08]$, and $\sigma_w = 0.004$ rad; during evaluation, the corresponding values are $\sigma_\phi = 0.015$ rad, $[-0.05, 0.05]$, and $\sigma_w = 0.005$ rad.

UPAIR uses a feature width 96, four prompted axial blocks, four attention heads, MLP expansion ratio 2, dropout rate 0.05, and correction scale $\alpha_{\text{res}} = 0.35$. Training draws the signal-to-noise ratio (SNR or E_b/N_0) uniformly from 0 to 18 dB for 5000 iterations with batch size 48. Evaluation uses batch size 96, and early stopping after 100 observed block errors.

Figures 4 and 5 summarize the BLER and NMSE behavior across the three evaluated scenarios. The middle panel in each figure corresponds to the mild phase-distortion scenario, which is the most informative setting because it combines the original DMRS density with residual symbol-wise phase rotation.

TABLE I
Scenario definitions used in the three evaluated settings.

Scenario	Phase Error	Add. DMRS	Description
Clean reference	Off	0	Reference case without phase impairment
Mild phase distortion	On	0	Main phase-impaired case
DMRS-rich mild phase	On	1	Same phase impairment with denser pilot support

The middle panel of Fig. 4, plots the left-most *practical* waterfall. Specifically, UPAIR reaches a BLER 10^{-2} at an SNR of about 5.46 dB. The best classical BLER comparator at that point is DD-CPE + LS at about 6.63 dB, while LS + 2D LMMSE requires about 6.66 dB. Hence the operating-point gain of UPAIR is about 1.18 dB relative to DD-CPE + LS and about 1.20 dB relative to LS + 2D LMMSE.

The perfect-CSI benchmark reaches the same BLER at about 4.91 dB, so the remaining gap with UPAIR is about 0.55 dB. In the middle panel of Fig. 5, UPAIR achieves the lowest NMSE across the evaluated E_b/N_0 range. LS + 2D LMMSE is the strongest classical estimator, but its average NMSE over 0–14 dB is still about $2.05\times$ higher, with the gap growing from about $1.34\times$ at 0 dB to about $3.49\times$ at 14 dB.

The three BLER panels of Fig. 4 also show that the gain of UPAIR is consistent. In the clean reference panel (no phase distortion), UPAIR reaches a BLER of 10^{-2} at an SNR pf about 5.45 dB, whereas the best classical receiver, LS + 2D LMMSE, needs about 6.25 dB, i.e., a gain of about 0.80 dB. The main phase-distortion panel (middle panel) gives almost the same UPAIR operating point, 5.46 dB, suggesting that the slot prompt and FiLM-conditioned refinement compensate for most of the mild phase error.

In the DMRS-rich panel, an additional DMRS symbol is inserted, providing pilot observations at two time locations instead of one. As a result, LS + 2D LMMSE becomes the strongest classical receiver for both BLER and NMSE, since the additional pilot symbol provides the second reference needed by LMMSE time interpolation to resolve symbol-wise phase drift more accurately. At a BLER of 10^{-2}, LS + 2D LMMSE requires about 5.44 dB SNR and UPAIR requires about 5.46 dB, i.e., the two curves are essentially tied.

Figure 5 shows the same trend across all three scenarios. Averaged over the evaluated E_b/N_0 grid, the best classical estimate is about $2.11\times$, $2.05\times$, and $1.47\times$ worse than UPAIR in the clean, mild phase-distortion, and DMRS-rich scenarios, respectively. The smaller reduction in the DMRS-rich case reflects the added DMRS support. Still, UPAIR providves the lowest NMSE at every evaluated E_b/N_0 point. This suggests that the learned front-end is most beneficial when pilot support is sparse or mild phase distortion is present.

VI. Conclusion

UPAIR is a prompted channel-estimation front-end that leaves the rest of the NR uplink receiver unchanged. By forming a slot prompt through DMRS-masked averaging of the projected feature tensor followed by a small MLP, refining an LS reference with FiLM-conditioned axial blocks, and producing both a channel estimate and an uncertainty map, it improves receiver performance while retaining the standard downstream detector and decoder. Across the clean, mild phase-distortion, and DMRS-rich scenarios, UPAIR gives the best practical NMSE among LS, DD-CPE + LS, and LS + 2D LMMSE. In BLER, it improves the BLER=10^{-2} operating point by about 0.80 dB in the clean scenario and about 1.18 dB in the mild phase-distortion scenario, while in the DMRS-rich scenario it is essentially tied with LS + 2D LMMSE around BLER 10^{-2}. These results indicate that prompted slot-wise conditioning is most valuable when pilot support is relatively sparse or when mild phase distortion is present.

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